Grade 9 Chemistry

Solved Exercises - English Medium - Balochistan Textbook Board (BTBB)

Complete book: all 16 units, every unit-end Exercise (multiple choice, restricted response and extended response questions, plus project prompts and numerical problems where the book prints them).

Unit 1: Nature of Science in Chemistry

Unit 2: Matter

Unit 3: Atomic Structure

Unit 4: Chemical Bonding

Unit 5: Stoichiometry

Unit 6: Electrochemistry

Unit 7: Chemical Energetics

Unit 8: Chemical Equilibrium

Unit 9: Acids and Bases

Unit 10: Periodic Table and Group Properties

Unit 11: Environmental Chemistry

Unit 12: Organic Chemistry

Unit 13: Biochemistry

Unit 14: Empirical Data Collection and Analysis

Unit 15: Separation Techniques

Unit 16: Qualitative Analysis

Source: BTBB Grade 9 Chemistry textbook, 208 pages. Medium confirmed from the book itself: cover, Contents (PDF p.4) and all units are in English. Exercises were read from rendered pages (book pp.6-7, 19-20, 33-34, 50-51, 62-63, 73-74, 82-83, 90-91, 99-100, 120-121, 144-145, 158-159, 168-169, 180-181, 189-190, 197-198). Questions are reproduced from each unit Exercise; answers are written in medium length, and numerical problems show every step. In-lesson Activity boxes are practical tasks, not question sets, and are not reproduced as questions. Book quirks noted honestly (answers follow the book's intended option; see the manifest for detail): Unit 3 MCQ i and vii ask the same thing (atomic number = protons). Unit 5 MCQ iv options (a) and (b) print the same value and MCQ vi mislabels its second option as (a) again; the answers are unaffected. Unit 6 MCQ iii has two defensible options (a reducing agent both loses electrons and reduces other substances); (b) is ticked. Unit 6 MCQ x omits the correct value: the oxidation state of Cl in HCl is -1, which is not among the printed options, so (b) 1 is ticked for its magnitude only. Unit 6 ERQ iv prints the thiosulphate species as Na2(S2O3)2; it is answered as sodium thiosulphate, Na2S2O3. Unit 15 repeats two MCQs (vii = ix, vi = x). Unit 16 prints its ninth MCQ as xi and its tenth as x. Unit 11 MCQ iii: agriculture and livestock farming is ticked as the main source of methane emissions (the main human source; wetlands are the largest natural source).

Unit 1: Nature of Science in Chemistry

Exercise 1

A. Multiple choice questions. (Correct option marked with a green tick.)

(1) Chemistry primarily focuses on the study of:

(a) living organisms

(b) matter and its properties ✓

(c) celestial bodies

(d) electrical engineering

(2) The option that is NOT a subfield of chemistry is:

(a) organic chemistry

(b) environmental Chemistry

(c) civil engineering ✓

(d) analytical chemistry

(3) The interdisciplinary field that integrates chemistry with living organisms is:

(a) polymer chemistry

(b) geochemistry

(c) biochemistry ✓

(d) medicinal chemistry

(4) The field of chemistry concerned with studying geological materials and celestial bodies is:

(a) geochemistry ✓

(b) medicinal chemistry

(c) analytical chemistry

(d) polymer chemistry

(5) The role that environmental chemistry plays in solving problems is:

(a) drug discovery

(b) pollution control and climate change ✓

(c) material science

(d) nuclear reactions

(6) The aspect that physical chemistry studies on the atomic and molecular level is:

(a) earth and environmental sciences

(b) behaviour of matter ✓

(c) organic compounds

(d) industrial chemicals

(7) Inorganic chemistry is concerned with the study of:

(a) living organisms

(b) organic compounds

(c) inorganic compounds ✓

(d) radioactive materials

(8) The study of radioactivity and nuclear reactions is known as:

(a) biochemistry

(b) environmental chemistry

(c) nuclear chemistry ✓

(d) polymer chemistry

(9) The subfield of chemistry that studies pollutants and their impact on air, water, and soil is:

(a) inorganic

(b) environmental ✓

(c) physical

(d) industrial

(10) Geochemistry mainly studies the chemical composition of:

(a) living tissues

(b) interstellar clouds

(c) geological materials like rocks and soil ✓

(d) plastics and synthetic polymers

B. Restricted Response Questions (RRQs).

(i) What does polymer chemistry study?

Polymer chemistry studies large molecules called polymers, which are made of many small repeating units joined together. It covers natural polymers such as proteins and cellulose, and synthetic polymers such as plastics, nylon and rubber, and how their structure gives them their properties.

(ii) Which branch of chemistry studies planets and stars?

Astrochemistry studies planets, stars and interstellar matter. It uses chemistry to explain what celestial bodies are made of, how molecules form in space, and how the chemical elements were produced in stars.

(iii) Which branch of chemistry helps us understand pollution?

Environmental chemistry helps us understand pollution. It studies pollutants in air, water and soil, where they come from, how they react and spread, and how they affect living things and the climate.

(iv) What question does analytical chemistry help answer?

Analytical chemistry helps answer the question: what is a sample made of, and how much of each substance does it contain? It identifies the substances present and measures their amounts accurately, for example in water testing or medicine checking.

(v) Write short notes on the following branches of chemistry: a) Biochemistry b) Environmental Chemistry c) Nuclear Chemistry

a) Biochemistry: the study of the chemical processes in living organisms, such as digestion, respiration and how enzymes and DNA work. b) Environmental Chemistry: the study of chemical changes in air, water and soil, including pollution and its control. c) Nuclear Chemistry: the study of radioactivity, nuclear reactions and the uses of radioisotopes in medicine, energy and industry.

C. Extended Response Questions (ERQs).

(i) Explain how chemistry helps us understand the earth (solids), air (gases), sea (liquids), and sky (plasmas) with suitable examples.

Chemistry explains the earth by studying solids such as rocks, minerals and soil, for example how limestone reacts with acids. It explains air by studying gases such as oxygen, nitrogen and carbon dioxide, for example how oxygen supports burning. It explains the sea by studying liquids such as water and salt solutions, for example how salts stay dissolved in seawater. It explains the sky by studying plasma, the high-energy state of matter found in the sun and stars and in lightning, where gases are ionised at very high temperatures.

(ii) Describe the difference between subfields and interdisciplinary fields of chemistry. Give two examples of each.

Subfields are the main branches of chemistry itself, divided by the kind of matter studied, for example organic chemistry, which studies carbon compounds, and inorganic chemistry, which studies other compounds. Interdisciplinary fields combine chemistry with another science, for example biochemistry, which joins chemistry with biology, and geochemistry, which joins chemistry with geology. Subfields stay inside chemistry, while interdisciplinary fields apply chemistry across subjects.

(iii) Choose any four branches of chemistry and write one essential question that scientists working in each field try to answer.

Organic chemistry: how can we make new carbon compounds, such as medicines and plastics? Analytical chemistry: what exactly is present in this sample, and in what amount? Environmental chemistry: how do pollutants move through air and water, and how can they be reduced? Nuclear chemistry: how do radioactive nuclei change, and how can radiation be used safely? Each branch is driven by its own central question about matter and its changes.

(iv) Differentiate between science, technology, and engineering. Give one clear example of each from chemistry or physical sciences.

Science is the process of discovering new knowledge through observation and experiment, for example discovering that acids react with metals to give hydrogen gas. Technology is the application of scientific knowledge to make useful tools, for example a microscope or a pH meter built from scientific principles. Engineering is the application of science and

technology to design and build products that solve a problem, for example designing a water-treatment plant that removes pollutants from drinking water.

(v) Explain how science, technology, and engineering work together in solving problems. Use a real-life example to support your answer.

Science, technology and engineering work together when a problem is first understood scientifically, then tackled with tools, then solved with a designed product. For example, polluted drinking water is first studied by science, which identifies the harmful chemicals present. Technology provides testing instruments and filters, and engineering designs a treatment plant that cleans the water for a whole town. Without science there would be no understanding, without technology no tools, and without engineering no working solution.

Unit 2: Matter

Exercise 2

A. Multiple choice questions. (Correct option marked with a green tick.)

(1) The fundamental characteristic of matter is:

(a) colour

(b) mass and volume ✓

(c) temperature

(d) pressure

(2) The following is NOT considered matter from a scientific perspective:

(a) water

(b) sound ✓

(c) trees

(d) silver

(3) The state of matter characterised by having a definite shape and volume is:

(a) gas

(b) liquid

(c) solid ✓

(d) plasma

(4) A typical property of liquids is:

(a) definite shape

(b) definite structure

(c) high compressibility

(d) ability to diffuse slowly ✓

(5) The average kinetic energy of gas molecules is directly proportional to:

(a) pressure of the gas

(b) temperature of the gas ✓

(c) volume of the gas

(d) number of molecules

(6) The temperature at which a bose-Einstein condensate is typically formed is:

(a) room temperature

(b) close to absolute zero ✓

(c) boiling point

(d) freezing point

(7) The typical size range of particles in a colloid is:

(a) 0.001nm to 0.01nm

(b) 1nm to 1000nm ✓

(c) 0.1nm to 1nm

(d) 10nm to 100nm

(8) An example of an unsaturated solution is a solution:

(a) with dissolved solute at its maximum capacity

(b) with excess solute settled at the bottom

(c) that can dissolve more solute ✓

(d) with equal amounts of solute and solvent

(9) Fog, classified as a type of mixture, is:

(a) solution

(b) suspension

(c) colloid ✓

(d) compound

(10) Solids are considered incompressible because:

(a) they have high kinetic energy

(b) their particles are widely spaced

(c) their particles are tightly packed ✓

(d) they have a definite shape

B. Restricted Response Questions (RRQs).

(i) What is diffusion? Provide one everyday example.

Diffusion is the movement of particles from a region of higher concentration to a region of lower concentration until they are evenly spread. An everyday example is the smell of perfume spreading through a room, or ink spreading slowly through water without stirring.

(ii) Why are gases compressible, while solids are not?

Gases are compressible because their particles are far apart with large empty spaces between them, so squeezing reduces those spaces. Solids are not compressible because their particles are already tightly packed with almost no space between them, so there is nowhere for them to move closer.

(iii) List two differences between homogeneous and heterogeneous mixtures.

First, a homogeneous mixture is uniform throughout, like salt solution, while a heterogeneous mixture is not uniform, like sand in water. Second, the components of a homogeneous mixture cannot be seen separately, while in a heterogeneous mixture the different parts can often be seen and separated physically.

(iv) Why is milk considered a colloid instead of a solution?

Milk is considered a colloid because its tiny fat and protein particles are dispersed through water but are larger than the particles in a true solution. They do not settle out like a suspension, and they scatter light, which is why milk looks white and opaque rather than clear.

(v) Why do liquids flow, whereas solids do not?

Liquids flow because their particles are less tightly packed than those of solids and can slide past one another, so the liquid takes the shape of its container. Solids do not flow because their particles are held in fixed positions by strong forces and can only vibrate, so a solid keeps its own shape.

C. Extended Response Questions (ERQs).

(i) Differentiate solids, liquids, and gases based on their shape, volume, compressibility, and intermolecular forces.

Solids have a definite shape and volume, are almost incompressible, and have strong intermolecular forces holding particles in fixed positions. Liquids have a definite volume but no fixed shape, are slightly compressible, and have weaker forces that let particles slide past one another. Gases have neither definite shape nor volume, are highly compressible, and have very weak forces, so their particles move freely and fill any container.

(ii) What is a solution? Describe the properties of a true solution and explain the terms solute and solvent with examples.

A solution is a homogeneous mixture in which one substance is dissolved evenly in another. A true solution is clear, its particles are of molecular size, it does not scatter light, and its solute does not settle out. The solute is the substance dissolved, for example salt in salt water, and the solvent is the substance that dissolves it, for example the water itself. In sugar solution, sugar is the solute and water is the solvent.

(iii) Describe the effect of temperature on the solubility of solids and gases in liquids.

For most solids, solubility in liquids increases as temperature rises, because the extra heat helps break the solid apart and spread its particles; a saturated solution made hot can dissolve more solid. For gases, the opposite happens: solubility in liquids decreases as temperature rises, because the gas particles gain energy and escape from the liquid. That is why warm fizzy drinks lose their gas faster than cold ones.

(iv) Differentiate between element, compound, and mixture.

An element consists of only one type of atom, for example oxygen or iron, and cannot be broken down by chemical means. A compound is formed when two or more elements combine chemically in a fixed ratio, for example water, H_2O, and it has properties different from its elements. A mixture is formed by physically combining substances in any ratio, for example air or salt solution, and its components keep their own properties and can be separated physically.

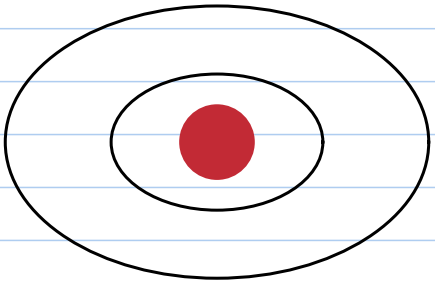
(v) What are colloids? List their characteristics and provide four examples from daily life.

Colloids are mixtures in which tiny particles, between 1 nm and 1000 nm in size, are dispersed through another substance. They do not settle out, they cannot be separated by ordinary filtration, and they scatter light. Four daily-life examples are milk, fog, jelly and

mayonnaise. Smoke in air and butter are further examples of colloids.

Unit 3: Atomic Structure

Diagram: Bohr model of a carbon atom (Unit 3)



Nucleus: 6 protons + 6 neutrons

Shells: 2 electrons + 4 electrons

Exercise 3

A. Multiple choice questions. (Correct option marked with a green tick.)

(1) The atomic number (Z) of an element is defined as the number of:

(a) neutrons in the nucleus

(b) electrons in all shells

(c) protons in the nucleus ✓

(d) protons and neutrons together

(2) The relative atomic mass of an element is based on:

(a) mass of the lightest isotope of the element

(b) average mass of isotopes compared with 1/12th mass of a carbon-12 atom ✓

(c) total number of protons and neutrons in the nucleus

(d) mass of a single atom measured in kilograms

(3) The charge of a proton is:

(a) negative

(b) positive ✓

(c) neutral

(d) variable

(4) The atomic model in which electrons are arranged in specific orbits or shells is:

(a) thomson's plum pudding model

(b) rutherford's atomic model

(c) bohr's atomic model ✓

(d) quantum mechanical model

(5) The force responsible for binding protons and neutrons together in the nucleus is:

(a) electromagnetic force

(b) gravitational force

(c) strong nuclear force ✓

(d) weak nuclear force

(6) A neutron has a charge that is:

(a) positive

(b) negative

(c) neutral ✓

(d) variable

(7) The atomic number of an element is equal to the number of:

(a) protons ✓

(b) neutrons

(c) electrons

(d) nucleons

(8) The mass of a proton compared to that of an electron is:

(a) 1/1836

(b) 1836 ✓

(c) 1839

(d) 1842

(9) An electron carries a charge equal in magnitude to that of a proton but opposite in sign. Its charge is:

(a) +1

(b) -1 ✓

(c) 0

(d) 1/1837

(10) The mass number of an element is:

(a) Number of protons

(b) Number of neutrons

(c) Sum of protons and electrons

(d) Sum of protons and neutrons ✓

B. Restricted Response Questions (RRQs).

(i) What does relative atomic mass mean?

Relative atomic mass is the average mass of the atoms of an element compared with one-twelfth of the mass of a carbon-12 atom. It takes account of all the isotopes of the element and how common each one is, so it is usually not a whole number, for example chlorine has a relative atomic mass of about 35.5.

(ii) Why do atoms have no overall charge?

Atoms have no overall charge because they contain equal numbers of positively charged

protons in the nucleus and negatively charged electrons around it. The positive and negative charges cancel each other exactly, so the atom as a whole is electrically neutral.

(iii) Briefly explain why a sodium ion has a charge of 1+ while a chloride ion has a charge of 1-.

A sodium atom has one electron in its outer shell, which it loses to reach a stable arrangement, leaving one more proton than electrons, so the ion is Na+ with a 1+ charge. A chlorine atom has seven outer electrons and gains one electron to complete its shell, giving one more electron than protons, so the ion is Cl- with a 1- charge.

(iv) Why do scientists conclude that electrons revolve around the nucleus and not the neutron?

Scientists conclude that electrons move around the nucleus because the nucleus, with its positively charged protons, attracts the negatively charged electrons and holds them in shells around it. A neutron is itself only a neutral particle inside the nucleus, not a centre of attraction, so it cannot be what the electrons orbit.

(v) How does the mass of a proton compare to that of an electron?

A proton is about 1836 times heavier than an electron. The proton's mass is about 1.673 x 10^-27 kg, while the electron's mass is only about 9.11 x 10^-31 kg, so the electron contributes almost nothing to the mass of the atom.

(vi) Some chemical elements have symbols derived from their Latin names. Identify the element for each given Latin name below: a) Natrium b) Kalium c) Ferrum d) Plumbum f) Aurum. Provide the modern English name and the chemical symbol for each.

a) Natrium: sodium, Na. b) Kalium: potassium, K. c) Ferrum: iron, Fe. d) Plumbum: lead, Pb. f) Aurum: gold, Au. (The book prints the list as a, b, c, d, f, skipping e.)

C. Extended Response Questions (ERQs).

(i) Explain the structure of the atom, including its nucleus and electron shells.

An atom has a tiny, dense, positively charged nucleus at its centre, containing protons and neutrons held together by the strong nuclear force. Around the nucleus, negatively charged electrons occupy shells or energy levels, filling the first shell with up to 2 electrons, the second with up to 8, and so on. Most of the atom is empty space, and almost all its mass is in the nucleus, while its chemical behaviour is decided by the electrons in the outermost shell.

(ii) What are isotopes, and how do they differ in terms of protons and neutrons?

Isotopes are atoms of the same element that have the same number of protons but different numbers of neutrons, so they have the same atomic number but different mass numbers. For example, carbon-12 has 6 protons and 6 neutrons, while carbon-14 has 6 protons and 8 neutrons. Because their electron arrangement is the same, isotopes have almost identical chemical properties, but different physical properties such as mass and

stability.

(iii) What is relative atomic mass, and how is it calculated from isotopes?

Relative atomic mass is the weighted average mass of an element's atoms compared with one-twelfth of a carbon-12 atom. It is calculated by multiplying the mass of each isotope by its percentage abundance, adding the results, and dividing by 100. For example, chlorine is about 75% chlorine-35 and 25% chlorine-37, giving (35 x 75 + 37 x 25) / 100 = 35.5, which is why chlorine's relative atomic mass is 35.5.

(iv) How can isotopes be used in practical applications such as carbon dating and medical imaging?

Carbon dating uses carbon-14, a radioactive isotope found in all living things, which decays at a known rate after death; measuring how much carbon-14 remains tells how old a fossil or artefact is. In medical imaging, radioactive isotopes such as technetium-99m are injected or swallowed, and the radiation they give off is detected to make images of organs, showing how they are working without surgery. Isotopes are also used to treat some cancers and to trace chemical reactions.

D. Project (activity prompt with an approach note).

(i) Project: Construct a simple 3D model of a carbon atom using coloured balls to represent protons, neutrons, and electrons, and label the nucleus and electron shells.

Approach: use six balls of one colour for the 6 protons and six of a second colour for the 6 neutrons, packed together as the nucleus, and six balls of a third colour for electrons, with two on an inner ring and four on an outer ring, then label the nucleus and the two electron shells.

Unit 4: Chemical Bonding

Exercise 4

A. Multiple choice questions. (Correct option marked with a green tick.)

(1) The primary reason atoms form chemical bonds is to:

(a) become unstable

(b) increase their atomic number

(c) achieve a stable electronic configuration ✓

(d) become radioactive

(2) The rule that indicates atoms generally have eight electrons in their outermost shell is:

(a) octet rule ✓

(b) duplet rule

(c) triplet rule

(d) hexlet rule

(3) The group of elements least likely to form chemical bonds because of their filled valence shells are:

(a) halogens

(b) noble gases ✓

(c) alkali metals

(d) transition metals

(4) The charge on a cation is:

(a) positive ✓

(b) negative

(c) neutral

(d) variable

(5) The type of bond formed by the transfer of electrons between atoms with a significant difference in electronegativity is:

(a) covalent bond

(b) metallic bond

(c) ionic bond ✓

(d) coordinate covalent Bond

(6) The state in which ionic compounds typically exist at room temperature is:

(a) gas

(b) liquid

(c) solid ✓

(d) plasma

(7) The property of metals that makes them good conductors of electricity is:

(a) high electronegativity

(b) low melting point

(c) delocalized electrons ✓

(d) covalent bonding

(8) The type of bond formed by the equal or mutual sharing of electrons between two atoms is:

(a) ionic bond

(b) metallic bond

(c) covalent bond ✓

(d) hydrogen bond

(9) The primary industrial use of graphite is:

(a) construction material

(b) electrical conductor in electrodes ✓

(c) fertiliser additive

(d) food preservative

(10) Metals are widely used in electrical wiring due to their:

(a) high melting point

(b) electrical conductivity ✓

(c) shiny appearance

(d) high density

B. Restricted Response Questions (RRQs).

(i) Why do representative elements in groups VA-VIIA tend to gain electrons?

Elements in groups VA-VIIA have five, six or seven electrons in their outer shell, so they are only one, two or three electrons short of a stable octet. Gaining those few electrons is much easier than losing five or more, so they tend to gain electrons and form negative ions.

(ii) Provide examples of substances that can ionise when dissolved in water.

Substances that ionise in water include sodium chloride, which gives Na+ and Cl- ions, and hydrogen chloride, which gives H+ and Cl- ions. Acids such as sulphuric acid and alkalis such as sodium hydroxide also ionise in water, producing ions that let the solution conduct electricity.

(iii) Explain why the bond in an oxygen molecule is called a double bond.

The bond in an oxygen molecule, O2, is called a double bond because the two oxygen atoms share two pairs of electrons between them. Each oxygen atom needs two more electrons to complete its octet, and by sharing two pairs each atom gains the two electrons it needs, giving a double covalent bond, O=O.

(iv) Differentiate between single, double, and triple covalent bonds.

A single covalent bond shares one pair of electrons between two atoms, as in H2 or HCl. A

double covalent bond shares two pairs, as in O2, making the bond shorter and stronger. A triple covalent bond shares three pairs, as in N2, making it even shorter and stronger. More shared pairs mean a stronger bond that needs more energy to break.

(v) Why do noble gases show low reactivity?

Noble gases show low reactivity because their outer shells are already complete, with eight electrons, or two in the case of helium. They already have the stable electronic configuration that other atoms bond to achieve, so they have no tendency to gain, lose or share electrons.

(vi) Solid sodium chloride does not conduct electricity, whereas a solution of sodium chloride does. why?

Solid sodium chloride does not conduct electricity because its Na+ and Cl- ions are held fixed in a crystal lattice and cannot move to carry charge. When it dissolves in water, the lattice breaks apart and the ions become free to move through the solution, so the solution conducts electricity.

C. Extended Response Questions (ERQs).

(i) What are the octet and duplet rules, and how do they relate to the stability of noble gas electronic configurations?

The octet rule states that atoms tend to gain, lose or share electrons until their outer shell has eight electrons, the arrangement of noble gases except helium. The duplet rule is the same idea for the first shell, which holds only two electrons, as in helium and in hydrogen when it forms H2 or loses its electron. Noble gases are stable precisely because they already satisfy these rules, with complete outer shells, so other atoms react in ways that copy a noble gas configuration.

(ii) How are ions formed? Explain the difference between cations and anions, providing examples.

Ions are formed when an atom gains or loses electrons, so that the numbers of protons and electrons are no longer equal. A cation is formed when an atom loses electrons, giving a positive ion, for example Na+ from sodium or Ca2+ from calcium. An anion is formed when an atom gains electrons, giving a negative ion, for example Cl- from chlorine or O2- from oxygen. Metals usually form cations and non-metals usually form anions.

(iii) Compare ionic, covalent, and metallic bonds with respect to their formation and key properties.

An ionic bond forms by transfer of electrons from a metal to a non-metal, giving oppositely charged ions held by electrostatic attraction; ionic compounds are usually hard solids with high melting points that conduct when molten or dissolved. A covalent bond forms by sharing electron pairs between non-metals; covalent substances may be gases, liquids or soft solids with low melting points and they usually do not conduct. A metallic bond forms when metal atoms release their outer electrons into a shared sea of delocalized electrons;

metals are shiny, malleable, ductile and good conductors of heat and electricity.

(iv) Why do ionic compounds possess high melting and boiling points? What factors influence these properties?

Ionic compounds have high melting and boiling points because their oppositely charged ions are held in a strong three-dimensional lattice by powerful electrostatic forces, and a large amount of heat energy is needed to break these forces. The melting point is higher when the ions are smaller and more highly charged, because the attraction is stronger, and lower when the ions are large or singly charged, because the attraction is weaker.

(v) What are the industrial applications of graphite, diamond, and metals? Explain how their properties are suited to these applications.

Graphite is used for electrodes and pencil leads because its layers slide easily and its delocalized electrons conduct electricity. Diamond is used for cutting tools and drill bits because it is the hardest known natural substance, with every carbon atom strongly bonded to four others. Metals are used for wiring, machinery and construction because they are strong, malleable, ductile and excellent conductors of heat and electricity, so each material's bonding and structure suit it to its job.

Unit 5: Stoichiometry

Exercise 5

A. Multiple choice questions. (Correct option marked with a green tick.)

(1) Stoichiometry deals with:

(a) colour changes

(b) physical properties

(c) quantitative relationships ✓

(d) speed of reaction

(2) The molecular formula of glucose is:

(a) C6H6

(b) C6H12O6 ✓

(c) C6H12

(d) CO2

(3) The type of formula that shows the simplest whole-number ratio of different atoms in a compound is:

(a) molecular formula

(b) empirical formula ✓

(c) chemical formula

(d) structural formula

(4) The number of particles present in ten mole of any substance is:

(a) 6.022 x 10^23

(b) 6.022 x 10^23

(c) 6.022 x 10^24 ✓

(d) 6.022 x 10^25

(5) The symbol that represents a solid state in a chemical equation is:

(a) (s) ✓

(b) (l)

(c) (g)

(d) (aq)

(6) If a sample of copper has a mass of 31.8 grams and contains 0.5 moles, its molar mass is:

(a) 15.9 g/mol

(b) 31.8 g/mol

(c) 63.6 g/mol ✓

(d) 0.0159 g/mol

(7) In the formula CaF2, the charge on the fluoride ions is:

(a) +1

(b) -1 ✓

(c) +2

(d) -2

(8) The molar mass of water (H2O) is:

(a) 18 g/mol ✓

(b) 22.414 dm3/mol

(c) 6.022 x 10^23 g/mol

(d) 2 g/mol

(9) The term that represents the actual number of atoms in a molecule is:

(a) Empirical formula

(b) Molecular formula ✓

(c) Chemical formula

(d) Structural formula

(10) The chemical formula for a compound composed of magnesium ions (Mg2+) and chloride ions (Cl-) is:

(a) MgCl2 ✓

(b) MgCl

(c) Mg2Cl2

(d) Mg2Cl

B. Restricted Response Questions (RRQs).

(i) Define the term Stoichiometry.

Stoichiometry is the branch of chemistry that deals with the quantitative relationships between reactants and products in a chemical reaction. It uses balanced equations to work out how much of each substance reacts and how much product forms.

(ii) What is the chemical formula for butane, and how many hydrogen atoms does it contain?

The chemical formula for butane is C4H10. It contains 10 hydrogen atoms, shown by the subscript 10 after H, together with 4 carbon atoms.

(iii) Give the empirical formula of glucose (C6H12O6).

The empirical formula of glucose is CH2O. It is found by dividing each subscript in C6H12O6 by their highest common factor, 6, giving C1H2O1, written as CH2O, the simplest whole-number ratio of its atoms.

(iv) State the difference between the terms 'mole' and 'molar mass'.

A mole is a counting unit for particles: one mole of any substance contains 6.022 x 10^23 particles. Molar mass is the mass of one mole of a substance, expressed in grams per mole, for example the molar mass of water is 18 g/mol. A mole counts particles, while molar mass weighs one mole of them.

(v) Calculate the molar mass of hydrogen sulphide (H2S).

Step 1: atomic mass of hydrogen = 1, and there are 2 hydrogen atoms: 2 x 1 = 2. Step 2: atomic mass of sulphur = 32, and there is 1 sulphur atom: 1 x 32 = 32. Step 3: add them: 2 + 32 = 34. The molar mass of hydrogen sulphide is 34 g/mol.

(vi) How many molecules are present in 1.5 moles of ammonia?

Step 1: one mole contains 6.022 x 10^23 molecules. Step 2: multiply by the number of moles: 1.5 x 6.022 x 10^23. Step 3: 1.5 x 6.022 = 9.033. There are 9.033 x 10^23 molecules of ammonia in 1.5 moles.

(vii) What information does a balanced chemical equation provide?

A balanced chemical equation shows the reactants and products of a reaction, their formulae, their physical states, and the mole ratio in which they react. It shows that atoms are conserved, so the number of each type of atom is the same on both sides, and it lets us calculate the amounts of reactants needed and products formed.

C. Extended Response Questions (ERQs).

(i) Discuss the significance of stoichiometry in chemical reactions, providing examples.

Stoichiometry matters because it turns a balanced equation into exact quantities for industry, medicine and the laboratory. For example, in making ammonia by the Haber process, stoichiometry tells how much nitrogen and hydrogen are needed and how much ammonia will form, so nothing is wasted. In a car airbag, it tells how much sodium azide must decompose to release enough nitrogen gas in a fraction of a second. Without stoichiometry, reactions would be guesswork, with wasted chemicals, low yields and even dangerous excesses.

(ii) Define the empirical formula of a compound with examples and explain how to determine it from a given molecular formula.

The empirical formula shows the simplest whole-number ratio of the atoms in a compound, for example CH2O for glucose and CH for benzene, whose molecular formula is C6H6. To determine it from a molecular formula, divide every subscript by the highest number that divides them all: for glucose C6H12O6, divide by 6 to get CH2O; for hydrogen peroxide H2O2, divide by 2 to get HO. If the subscripts have no common divisor, as in water H2O, the empirical and molecular formulae are the same.

(iii) Define a mole. Explain its significance in chemistry, and relate it to Avogadro's number.

A mole is the amount of a substance that contains exactly 6.022 x 10^23 particles, which may be atoms, molecules, ions or formula units. This number is Avogadro's number, named after Amedeo Avogadro. The mole is significant because it links the invisible world of atoms to measurable masses: one mole of carbon weighs 12 g, one mole of water weighs 18 g, so chemists can weigh out exact numbers of particles on a balance and use balanced

equations to predict reaction quantities.

D. Numerical Problems (every step shown).

(i) Calculate the number of moles in 39.2g of H2SO4 (molar mass = 98g/mol).

Step 1: use the formula: number of moles = mass / molar mass. Step 2: substitute the values: 39.2 / 98. Step 3: divide: 39.2 / 98 = 0.4. There are 0.4 moles of H2SO4 in 39.2 g.

(ii) A sample of copper has a mass of 15.9g and contains 0.25moles. What is its molar mass?

Step 1: use the formula: molar mass = mass / number of moles. Step 2: substitute the values: 15.9 / 0.25. Step 3: divide: 15.9 / 0.25 = 63.6. The molar mass of copper is 63.6 g/mol.

(iii) Calculate the relative molecular mass of sulphuric acid (H2SO4).

Step 1: hydrogen: 2 atoms x 1 = 2. Step 2: sulphur: 1 atom x 32 = 32. Step 3: oxygen: 4 atoms x 16 = 64. Step 4: add them: 2 + 32 + 64 = 98. The relative molecular mass of sulphuric acid is 98.

(iv) Determine the relative formula mass of aluminium sulphate (Al2(SO4)3).

Step 1: aluminium: 2 atoms x 27 = 54. Step 2: one sulphate group SO4 = 32 + (4 x 16) = 96, and there are 3 groups: 3 x 96 = 288. Step 3: add them: 54 + 288 = 342. The relative formula mass of aluminium sulphate is 342.

(v) If you have 2.5moles of sodium bromide, determine the number of formula units in this sample.

Step 1: one mole contains 6.022 x 10^23 formula units. Step 2: multiply by the number of moles: 2.5 x 6.022 x 10^23. Step 3: 2.5 x 6.022 = 15.055, so 15.055 x 10^23 = 1.5055 x 10^24. There are 1.5055 x 10^24 formula units of sodium bromide in 2.5 moles.

Unit 6: Electrochemistry

Exercise 6

A. Multiple choice questions. (Correct option marked with a green tick.)

(1) In a redox reaction, the process that involves the gain of electrons is:

(a) oxidation

(b) reduction ✓

(c) both oxidation and reduction

(d) neither

(2) The oxidation state of lead in PbO_2 is:

(a) 1+

(b) 2+

(c) 3+

(d) 4+ ✓

(3) The role of a reducing agent in a chemical reaction is:

(a) gains electrons

(b) loses electrons ✓

(c) oxidises other substances

(d) reduces other substances

(4) The oxidation number of sodium in its free state is:

(a) 1

(b) 0 ✓

(c) 1+

(d) 2+

(5) The sum of oxidation numbers in a neutral compound equals:

(a) 1

(b) 0 ✓

(c) 1+

(d) 2+

(6) The compound that acts as a strong oxidising agent is:

(a) hydrogen sulphide (H_2S)

(b) potassium dichromate ($K_2Cr_2O_7$) ✓

(c) sodium chloride (NaCl)

(d) iron(II) oxide (FeO)

(7) The colour change observed when potassium manganate(VII) reacts in a redox reaction is:

(a) purple to brown ✓

(b) brown to purple

(c) blue to green

(d) colourless to red

(8) The formula of an ionic compound can be derived by:

(a) a random combination of ions

(b) balancing the charges of cations and anions ✓

(c) adding hydrogen to the compound

(d) removing oxygen from the compound

(9) The method of corrosion prevention that involves attaching a more reactive metal as an anode to a less reactive metal surface is:

(a) painting

(b) galvanising

(c) sacrificial coating ✓

(d) oil or grease coating

(10) The oxidation state of chlorine in HCl is:

(a) 1+

(b) 1 ✓

(c) 0

(d) 2+

B. Restricted Response Questions (RRQs).

(i) Define the oxidation number and state the oxidation number of oxygen in carbon dioxide.

The oxidation number is the charge an atom appears to have in a compound, worked out by fixed rules, such as oxygen usually being -2 and hydrogen +1. In carbon dioxide, CO2, oxygen has an oxidation number of -2, so the two oxygen atoms total -4 and carbon must be +4 for the neutral molecule to sum to zero.

(ii) Summarise the sum of oxidation numbers in a neutral compound and provide an example to illustrate this concept.

In a neutral compound, the oxidation numbers of all the atoms add up to zero. For example, in water, H2O, the two hydrogen atoms are +1 each, giving +2, and oxygen is -2, so +2 + (-2) = 0. In a polyatomic ion, by contrast, they add up to the charge on the ion.

(iii) Potassium and chlorine react to form potassium chloride. Is it a redox reaction? If so, why?

Yes, it is a redox reaction. Potassium is oxidised because it loses one electron, going from oxidation number 0 to +1 in K+. Chlorine is reduced because each chlorine atom gains one electron, going from 0 to -1 in Cl-. Because oxidation and reduction happen together, the reaction 2K + Cl2 -> 2KCl is a redox reaction.

(iv) Identify the oxidising and reducing agents in the reactions given below by

observing the gain and loss of oxygen. 2Mg + O2 -> 2MgO; Fe + CuSO4 -> FeSO4 + Cu

In 2Mg + O2 -> 2MgO, magnesium gains oxygen, so magnesium is oxidised and is the reducing agent, while oxygen is the oxidising agent because it causes the oxidation. In Fe + CuSO4 -> FeSO4 + Cu, iron displaces copper: iron is oxidised from 0 to +2 and is the reducing agent, while Cu2+ is reduced to copper metal, so copper sulphate is the oxidising agent.

(v) Briefly discuss Roman numerals for oxidation states in compounds. Give examples of elements with variable oxidation states and their Roman numerals.

Roman numerals in a compound name show the oxidation state of an element that can have more than one, so that the formula is clear. For example, iron(II) oxide is FeO, where iron is +2, while iron(III) oxide is Fe2O3, where iron is +3. Copper(I) oxide is Cu2O and copper(II) oxide is CuO, and lead(II) oxide is PbO while lead(IV) oxide is PbO2.

(vi) Describe the rusting process of iron, including the chemical reactions involved.

Rusting is the corrosion of iron in the presence of water and oxygen. Iron is oxidised to iron(II) ions: Fe -> Fe2+ + 2e-. Oxygen from the air is reduced with water: O2 + 2H2O + 4e- -> 4OH-. The Fe2+ and OH- ions combine to form iron(II) hydroxide, which is further oxidised by air to hydrated iron(III) oxide, Fe2O3.xH2O, the brown flaky substance we call rust. Salt water speeds the process up.

C. Extended Response Questions (ERQs).

(i) Describe the three definitions of oxidation and reduction in electrochemistry. Provide examples for each definition.

First, in terms of oxygen: oxidation is gain of oxygen, as when magnesium burns to magnesium oxide, and reduction is loss of oxygen, as when copper(II) oxide is reduced to copper by hydrogen. Second, in terms of hydrogen: oxidation is loss of hydrogen, as when hydrogen sulphide is oxidised to sulphur, and reduction is gain of hydrogen, as when an alkene gains hydrogen to become an alkane. Third, in terms of electrons: oxidation is loss of electrons, as when Fe2+ loses an electron to become Fe3+, and reduction is gain of electrons, as when Cu2+ gains two electrons to become copper metal.

(ii) Explore methods to prevent corrosion, including alloying, oil or grease coating, paint coating, galvanising, and sacrificial coating.

Alloying mixes iron with other metals, as in stainless steel, where chromium forms a protective oxide layer that stops rust. Oil or grease coating and paint coating keep water and oxygen away from the metal surface, but they fail if scratched. Galvanising coats iron with a layer of zinc, which protects it even when scratched because zinc corrodes first. Sacrificial coating, or cathodic protection, attaches a more reactive metal such as magnesium or zinc to the structure, so that metal corrodes instead of the iron, as used on

ships and pipelines.

(iii) Explain how the formula of an ionic compound is derived from the charges of its cations and anions. Provide a step-by-step example using the reaction between calcium ions (Ca2+) and phosphate ions (PO4 3-), including the identification of ions and the balancing of charges.

Step 1: identify the ions: calcium is Ca2+ and phosphate is PO4 3-. Step 2: the total positive charge must equal the total negative charge. Step 3: three Ca2+ ions give +6, and two PO4 3- ions give -6, so +6 + (-6) = 0. Step 4: write the formula with those numbers as subscripts: Ca3(PO4)2. The brackets show that the subscript 2 applies to the whole phosphate group, giving calcium phosphate.

(iv) Determine the oxidation state of 'S' in the following species: (a) H2S (b) SO2 (c) SO3 (d) Na2(S2O3)2

Step 1, H2S: hydrogen is +1, so 2(+1) + S = 0, giving S = -2. Step 2, SO2: oxygen is -2, so S + 2(-2) = 0, giving S = +4. Step 3, SO3: S + 3(-2) = 0, giving S = +6. Step 4, the thiosulphate species: in sodium thiosulphate, Na2S2O3, 2(+1) + 2S + 3(-2) = 0, giving an average oxidation state of +2 for sulphur (in fact the two sulphur atoms are -2 and +6). Note: the book prints this species as Na2(S2O3)2.

D. Project (activity prompt with an approach note).

(i) Project: Examine corrosion prevention by placing four iron nails in saltwater: one bare, one painted, one coated with oil, and one wrapped with zinc. Observe rusting over a week and compare which method best protects the iron.

Approach: put each nail in a separate jar of saltwater, keep them in the same place, and record the rust on each nail daily for a week. The zinc-wrapped nail should stay almost rust-free because zinc corrodes in place of iron, the painted and oiled nails should rust little unless their coating is damaged, and the bare nail should rust most.

Unit 7: Chemical Energetics

Exercise 7

A. Multiple choice questions. (Correct option marked with a green tick.)

(1) The branch of thermodynamics that focuses on energy changes in chemical reactions is:

(a) chemical kinetics

(b) chemical energetics ✓

(c) physical chemistry

(d) biochemistry

(2) In an endothermic reaction, energy is absorbed:

(a) by the system from the surroundings ✓

(b) by the surroundings from the system

(c) in both directions between the system and the surroundings

(d) no net transfer of energy

(3) The enthalpy change (delta H) of a reaction is defined as:

(a) heat absorbed or released at constant pressure ✓

(b) variation in temperature of the system

(c) variation in pressure of the system

(d) change in mass of reactants or products

(4) The minimum amount of energy needed to start a chemical reaction is:

(a) enthalpy

(b) transition state

(c) energy change

(d) activation energy ✓

(5) The symbol delta represents in thermodynamics:

(a) mass of

(b) change in ✓

(c) rate of

(d) heat stored

(6) In an exothermic reaction, the enthalpy change (delta H) is always:

(a) zero

(b) positive

(c) negative ✓

(d) constant

(7) The type of system that exchanges both energy and matter with its surroundings is:

(a) closed system

(b) isolated system

(c) open system ✓

(d) dynamic system

(8) The role of a catalyst in a chemical reaction is to:

(a) increase activation energy

(b) decrease activation energy ✓

(c) produce more reactants

(d) inhibit product formation

(9) A negative enthalpy change (delta H < 0) signifies that the reaction is:

(a) Endothermic

(b) Exothermic ✓

(c) Isothermal

(d) Non-spontaneous

(10) The greater the activation energy:

(a) The lesser the energy needed to start a reaction

(b) The more energy needed to start a reaction ✓

(c) The lesser the energy needed to complete a reaction

(d) The more energy needed to complete a reaction

B. Restricted Response Questions (RRQs).

(i) What is the definition of a system in the context of thermodynamics?

In thermodynamics, a system is the part of the universe chosen for study, such as the reactants in a beaker, while everything else is the surroundings. The system and surroundings together make up the universe, and energy and sometimes matter can pass between them across the boundary.

(ii) Define exothermic reactions and give two examples.

Exothermic reactions release energy to the surroundings, usually as heat, so the surroundings get warmer and the enthalpy change is negative. Two examples are the combustion of natural gas in a cooker and the neutralisation of an acid with an alkali, both of which give out heat.

(iii) Define activation energy and briefly discuss its role in reaction kinetics.

Activation energy is the minimum energy that reactant particles must have for a collision to lead to a reaction. It acts as an energy barrier: particles with less energy simply bounce off each other, while those with at least the activation energy can break bonds and form products. The higher the activation energy, the slower the reaction, because fewer particles have enough energy.

(iv) Briefly explain why breaking bonds is an endothermic process.

Breaking bonds is endothermic because energy must be supplied to pull atoms apart

against the forces holding them together. No energy is released when a bond breaks; energy is only released later, when new bonds form in the products. That is why bond breaking always takes energy in.

(v) How can you use the energy from exothermic reactions?

The energy from exothermic reactions can be used for heating homes and cooking, as in burning natural gas or coal. It drives engines, where burning fuel releases heat that is turned into motion, and it runs power stations, where heat from combustion makes steam to drive turbines and generate electricity.

(vi) How do bond breaking and bond formation affect energy changes in reactions?

Bond breaking absorbs energy, while bond formation releases energy, and the overall enthalpy change is the difference between the two. If more energy is released in forming the new bonds than is absorbed in breaking the old ones, the reaction is exothermic. If breaking bonds takes more energy than forming new ones gives back, the reaction is endothermic.

(vii) Write a balanced chemical equation for the complete burning of coal.

Coal is mostly carbon, and its complete burning in oxygen gives carbon dioxide: C + O2 -> CO2. The equation is already balanced, with one carbon atom and two oxygen atoms on each side, and the reaction is strongly exothermic.

C. Extended Response Questions (ERQs).

(i) Define chemical energetics and explain its importance in understanding chemical reactions, with examples.

Chemical energetics is the branch of thermodynamics that studies the energy changes during chemical reactions, especially enthalpy changes. It is important because it tells us whether a reaction gives out or takes in heat, how much energy is involved, and whether a reaction is likely to happen. For example, it explains why burning fuel heats a home, why dissolving ammonium chloride cools water, and how much energy a power station can get from a tonne of coal.

(ii) Describe open, closed, and isolated systems. How do they differ in matter and energy exchange? Give real-life examples.

An open system exchanges both matter and energy with its surroundings, for example a boiling pot without a lid, where steam escapes and heat enters. A closed system exchanges energy but not matter, for example a sealed bottle of water being heated, where heat passes through the glass but no water escapes. An isolated system exchanges neither matter nor energy, for example a well-insulated thermos flask, which keeps a drink hot by almost completely blocking heat transfer.

(iii) How does activation energy differ in exothermic and endothermic reactions? Use examples and energy levels of reactants/products.

In an exothermic reaction, such as burning methane, the products are at a lower energy level than the reactants, and the activation energy is the energy hump the reactants must climb before falling to the lower product level, releasing the difference as heat. In an endothermic reaction, such as photosynthesis or dissolving ammonium nitrate, the products are at a higher energy level than the reactants, so the activation energy is larger and the system must keep absorbing energy to stay at the product level. In both cases the activation energy is the barrier between reactants and products, but the final energy level is lower for exothermic and higher for endothermic reactions.

(iv) How do catalysts and enzymes affect reaction pathways? Explain their effect on activation energy and reaction efficiency.

Catalysts and enzymes provide a different reaction pathway with a lower activation energy, so more reactant particles have enough energy to react and the reaction goes faster. A catalyst is not used up in the reaction and can be reused, for example iron in the Haber process. Enzymes are biological catalysts, such as amylase in saliva, which work under mild body conditions and are highly specific to one reaction. By lowering the barrier without changing the reactants or products, both make reactions faster and more efficient.

D. Project (activity prompt with an approach note).

(i) Project: Investigate the enthalpy changes of ammonium chloride and sodium hydroxide. Record the temperature change for each dissolution, classify them as endothermic or exothermic, and present your findings.

Approach: dissolve a measured mass of ammonium chloride in water and record the temperature fall, showing an endothermic change, then dissolve sodium hydroxide pellets in water and record the temperature rise, showing an exothermic change. Present both results in a table with the temperature change and the classification of each dissolution.

Unit 8: Chemical Equilibrium

Exercise 8

A. Multiple choice questions. (Correct option marked with a green tick.)

(1) The condition that defines chemical equilibrium is:

(a) forward rate > Reverse rate

(b) forward rate < Reverse rate

(c) forward rate = Reverse rate ✓

(d) forward rate + Reverse rate = 0

(2) For the reversible reaction N2 + 3H2 -> 2NH3, increasing the pressure will shift the equilibrium:

(a) towards reactants

(b) towards products ✓

(c) not affected

(d) stops the reaction

(3) A reaction is said to be at equilibrium when:

(a) the reaction is complete

(b) product concentration is zero

(c) forward and reverse rates are equal ✓

(d) the forward rate is maximum

(4) The removal of products from a system at equilibrium will:

(a) shift the equilibrium towards the reactants

(b) shift the equilibrium towards the products ✓

(c) keep the equilibrium unchanged

(d) stop the reaction completely

(5) A change in physical conditions causes the equilibrium to:

(a) remain unaffected

(b) shift entirely towards reactants

(c) shift entirely towards products

(d) shift towards either reactants or products ✓

(6) The type of equilibrium that exists in chemical reactions at the molecular level is:

(a) static equilibrium

(b) temporary equilibrium

(c) dynamic equilibrium ✓

(d) mechanical equilibrium

(7) The dissociation of PCl5 (PCl5 <-> PCl3 + Cl2) is favoured by:

(a) low temperature and high pressure

(b) high temperature and low pressure ✓

(c) low temperature and low pressure

(d) high temperature and high pressure

(8) The blue colour of hydrated copper(II) sulphate changes to white on heating because:

(a) water of crystallisation is released ✓

(b) oxygen escapes from the salt

(c) the salt decomposes to copper oxide

(d) sulphur dioxide gas is evolved

(9) When pressure increases in a gaseous system at equilibrium, it:

(a) always shifts right

(b) shifts to the side with more gas molecules

(c) shifts to the side with fewer gas molecules ✓

(d) no effect

(10) On hydration, anhydrous cobalt(II) chloride changes colour from:

(a) blue to white

(b) blue to pink ✓

(c) pink to white

(d) pink to blue

B. Restricted Response Questions (RRQs).

(i) What is meant by dynamic equilibrium?

Dynamic equilibrium is the state in a closed system where the forward and reverse reactions continue at equal rates, so the concentrations of reactants and products stay constant even though the reactions have not stopped. The word dynamic stresses that molecules keep reacting in both directions, unlike a static state where nothing happens.

(ii) Differentiate between static and dynamic equilibrium.

In static equilibrium, nothing is happening at all, for example a book resting on a table, with no movement or change. In dynamic equilibrium, the forward and reverse processes continue at equal rates, for example a sealed bottle of water where evaporation and condensation balance each other, so the water level stays constant while molecules keep moving between liquid and vapour.

(iii) How does increasing the concentration of a reactant affect equilibrium?

Increasing the concentration of a reactant makes the forward reaction faster, so the equilibrium shifts towards the products to use up the extra reactant. Once the system settles, the product concentration is higher than before, while the reactant concentration returns close to its original level, following Le Chatelier's principle.

(iv) What type of reaction is favoured when the temperature is decreased?

When the temperature is decreased, the exothermic reaction is favoured, because the system shifts in the direction that releases heat to oppose the cooling. For example, in the Haber process, lowering the temperature favours the forward, exothermic formation of ammonia, although in practice a compromise temperature is used so the reaction is not too slow.

(v) Explain why the concentration of products remains constant at equilibrium.

The concentration of products remains constant at equilibrium because products are being formed by the forward reaction at exactly the same rate as they are being used up by the reverse reaction. Formation and consumption balance each other, so the measured concentration does not change, even though both reactions are still running.

C. Extended Response Questions (ERQs).

(i) Describe reversible and irreversible reactions, providing examples for each type.

A reversible reaction can go in both directions, and is shown with a double arrow, for example the Haber process, N2 + 3H2 <-> 2NH3, or the dissociation of ammonium chloride into ammonia and hydrogen chloride. An irreversible reaction goes in one direction only, usually because a product escapes or the change cannot be undone, for example burning wood to ash and gases, or magnesium burning in air to magnesium oxide. Reversible reactions in closed systems can reach equilibrium, while irreversible reactions go to completion.

(ii) Explain why a closed system is essential for establishing equilibrium, using the example of vapour pressure in a volatile solvent.

A closed system is essential because equilibrium needs the forward and reverse processes to balance, which is impossible if a reactant or product escapes. For example, in a closed bottle of a volatile solvent such as ether, liquid evaporates until the vapour above it reaches a constant pressure, because evaporation and condensation then happen at equal rates. In an open bottle, the vapour escapes, evaporation never balances, and the liquid simply evaporates away without ever reaching equilibrium.

(iii) Describe the effect of temperature on the equilibrium of hydrated compounds, taking copper(II) sulphate as an example. How does heating affect the position of equilibrium?

Hydrated copper(II) sulphate, CuSO4.5H2O, is blue because of its water of crystallisation, and it exists in equilibrium with white anhydrous copper(II) sulphate and water: CuSO4.5H2O <-> CuSO4 + 5H2O. Heating drives off the water and shifts the equilibrium to the right, turning the crystals white, because the forward, dehydration step is endothermic and favoured by heat. Cooling in moist air reverses it, and the white solid turns blue again as it regains water.

Unit 9: Acids and Bases

Exercise 9

A. Multiple choice questions. (Correct option marked with a green tick.)

(1) According to the Bronsted-Lowry theory, an acid is a substance that:

(a) accepts protons

(b) donates hydroxide ions

(c) donates protons ✓

(d) accepts electrons

(2) The ion responsible for the acidic properties of a solution is:

(a) OH-

(b) Cl-

(c) H+ ✓

(d) Na+

(3) The best description of an alkali is a:

(a) base that is insoluble in water

(b) salt that reacts with water

(c) base that dissolves in water to produce OH- ions ✓

(d) base that dissolves in water to produce H+ ions

(4) A strong acid among the following is:

(a) CH3COOH

(b) H2CO3

(c) HCl ✓

(d) NH4OH

(5) A neutralisation reaction between an acid and a base typically produces:

(a) acid and gas

(b) salt and water ✓

(c) base and oxygen

(d) hydrogen gas only

(6) The following is NOT a property of acids:

(a) sour taste

(b) react with metals to produce hydrogen gas

(c) turn red litmus paper blue ✓

(d) pH less than 7

(7) When an acid reacts with a carbonate, it produces:

(a) salt and hydrogen gas

(b) salt, water, and carbon dioxide ✓

(c) only water

(d) salt and oxygen

(8) Sodium hydroxide (NaOH) is:

(a) weak base

(b) insoluble base

(c) strong alkali ✓

(d) weak acid

(9) The gas released when an acid reacts with a metal is:

(a) oxygen

(b) carbon dioxide

(c) hydrogen ✓

(d) nitrogen

(10) The best description of a neutralisation reaction is:

(a) acid reacts with metal to form hydrogen gas

(b) base reacts with water to produce hydroxide ions

(c) acid reacts with base to form salt and water ✓

(d) acid reacts with carbonate to produce carbon dioxide

B. Restricted Response Questions (RRQs).

(i) Which ion do alkalis release in aqueous solutions?

Alkalis release hydroxide ions, OH-, in aqueous solution. For example, sodium hydroxide dissolves in water to give Na+ and OH- ions, and it is the hydroxide ion that gives alkalis their characteristic properties, such as turning red litmus blue.

(ii) What is the pH value of a neutral solution?

The pH value of a neutral solution is 7 at room temperature. Pure water is neutral, with equal concentrations of H+ and OH- ions, and a pH below 7 is acidic while a pH above 7 is alkaline.

(iii) What is formed when an acid reacts with a carbonate?

When an acid reacts with a carbonate, a salt, water and carbon dioxide gas are formed. For example, hydrochloric acid reacts with calcium carbonate to give calcium chloride, water and carbon dioxide, which is seen as fizzing.

(iv) State one difference between a base and an alkali.

All alkalis are bases, but not all bases are alkalis: an alkali is a base that dissolves in water, while a base that does not dissolve in water is not an alkali. For example, sodium hydroxide is an alkali because it dissolves, while copper(II) oxide is a base that neutralises acids but does not dissolve in water.

(v) Why are strong acids good conductors of electricity in aqueous solutions?

Strong acids are good conductors because they dissociate completely in water into ions, giving a high concentration of mobile H+ ions and negative ions. These free ions carry

electric charge through the solution, so the more complete the dissociation, the better the conduction.

C. Extended Response Questions (ERQs).

(i) Explain the Bronsted-Lowry concept of acids and bases. Give one example of each.

In the Bronsted-Lowry concept, an acid is a proton donor and a base is a proton acceptor, where a proton is a hydrogen ion, H+. For example, in the reaction of hydrogen chloride with ammonia, HCl donates a proton and is the acid, while NH3 accepts the proton to become NH4+ and is the base. This concept is wider than the older idea of acids giving H+ in water, because it also covers reactions without water.

(ii) Differentiate between strong and weak acids with suitable reactions.

A strong acid dissociates completely in water, giving a high concentration of H+ ions, for example HCl -> H+ + Cl-, with almost every molecule split into ions. A weak acid dissociates only partly, so most molecules stay whole and few H+ ions are produced, for example CH3COOH <-> CH3COO- + H+, where the double arrow shows the incomplete, reversible dissociation. Strong acids therefore have lower pH and conduct better than weak acids of the same concentration.

(iii) State and explain three physical properties and two chemical properties of acids.

Three physical properties: acids have a sour taste, as in vinegar and citrus fruits; they are corrosive and can damage skin and materials; and their solutions conduct electricity because they contain ions. Two chemical properties: acids react with metals to release hydrogen gas, for example zinc with dilute sulphuric acid gives zinc sulphate and hydrogen; and acids react with bases in neutralisation to give a salt and water, for example HCl with NaOH gives NaCl and H2O. Acids also turn blue litmus paper red.

(iv) With the help of chemical equations, explain how acids react with metals, carbonates, and alkalis.

With metals: an acid reacts with a reactive metal to give a salt and hydrogen gas, for example 2HCl + Zn -> ZnCl2 + H2. With carbonates: an acid gives a salt, water and carbon dioxide, for example 2HCl + CaCO3 -> CaCl2 + H2O + CO2. With alkalis: an acid is neutralised to a salt and water, for example HCl + NaOH -> NaCl + H2O. In each case the acid provides H+ ions, which are replaced by the metal, released as gas with carbonate, or combined with OH- to form water.

Unit 10: Periodic Table and Group Properties

Exercise 10

A. Multiple choice questions. (Correct option marked with a green tick.)

(1) In the periodic table, the columns of elements are called:

(a) rows

(b) periods

(c) groups ✓

(d) families

(2) The elements in Group 17 are known as:

(a) alkali metals

(b) halogens ✓

(c) noble gases

(d) alkaline earth metals

(3) The number of groups present in the modern periodic table is:

(a) 12

(b) 14

(c) 18 ✓

(d) 20

(4) Elements in the same group of the periodic table have:

(a) the same atomic mass

(b) similar chemical properties ✓

(c) the exact number of neutrons

(d) different electron configurations

(5) Atomic radius as you move down a group:

(a) increases ✓

(b) decreases

(c) remains the same

(d) irregularly

(6) Ionisation energy as you move from left to right across a period:

(a) decreases

(b) increases ✓

(c) remains constant

(d) varies irregularly

(7) The property that transition metals typically show is:

(a) fixed oxidation states

(b) variable oxidation states ✓

(c) high reactivity

(d) non-metallic characteristics

(8) The group that typically forms anions with a charge of -1 is:

(a) group IA

(b) group IIA

(c) group VIIA ✓

(d) group IVA

(9) The alkali metal that reacts most explosively with water is:

(a) lithium

(b) sodium

(c) potassium

(d) caesium ✓

(10) Noble gases are unreactive because:

(a) they are non-metals

(b) they have low density

(c) they exist as diatomic molecules

(d) they have complete outer electron shells ✓

B. Restricted Response Questions (RRQs).

(i) Why does the atomic radius decrease from left to right across a period?

Across a period, each new element adds one proton to the nucleus and one electron to the same shell, so the nuclear charge pulling on the electrons grows stronger while the shielding stays almost the same. The electrons are therefore pulled closer to the nucleus, and the atomic radius decreases from left to right.

(ii) What are the criteria for classifying an element into the s-, p-, d-, or f-block?

An element is placed in a block according to which type of subshell its last, or valence, electron enters. If the last electron goes into an s subshell, the element is in the s-block; into a p subshell, the p-block; into a d subshell, the d-block; and into an f subshell, the f-block. For example, sodium is s-block, chlorine is p-block, iron is d-block and uranium is f-block.

(iii) Name three properties that distinguish metals from nonmetals.

First, metals are shiny, malleable and ductile, while non-metals are dull and brittle and cannot be drawn into wires. Second, metals are good conductors of heat and electricity, while non-metals, except graphite, are poor conductors. Third, metals tend to lose electrons and form positive ions, while non-metals tend to gain or share electrons and form negative ions or covalent compounds.

(iv) Explain why potassium is more reactive than lithium.

Potassium is more reactive than lithium because its single outer electron is in the fourth shell, much further from the nucleus than lithium's outer electron in the second shell. The

extra inner shells shield the outer electron from the nuclear pull, so potassium loses its electron more easily and reacts more vigorously, for example bursting into flame in water while lithium reacts gently.

(v) Describe one chemical property that distinguishes transition metals from alkali metals.

One distinguishing chemical property is that transition metals show variable oxidation states, while alkali metals show only +1. For example, iron forms both Fe2+ and Fe3+ compounds, while sodium forms only Na+ compounds, because transition metals can lose different numbers of d electrons as well as s electrons.

(vi) What type of ions do alkali metals form, and why?

Alkali metals form +1 ions, because each has a single electron in its outer shell, which it loses easily to reach a stable noble gas configuration. For example, sodium loses its one outer electron to become Na+, and potassium loses its one outer electron to become K+, leaving a complete shell underneath.

C. Extended Response Questions (ERQs).

(i) Describe how the electron configuration of an element determines its position in the periodic table.

The electron configuration decides both the period and the group of an element. The number of occupied shells gives the period: sodium, with electrons in three shells (2,8,1), is in period 3. The number of outer-shell electrons gives the group for main-group elements: sodium, with one outer electron, is in group 1, while chlorine, with seven outer electrons (2,8,7), is in group 17. The block follows from which subshell the last electron enters, so configuration alone fixes an element's address in the table.

(ii) Define ionisation energy and explain its trend across a period and down a group in the periodic table.

Ionisation energy is the energy needed to remove one electron from a gaseous atom, forming a positive ion. Across a period it increases, because the growing nuclear charge pulls electrons tighter and makes them harder to remove. Down a group it decreases, because each new shell puts the outer electron further from the nucleus and extra inner shells shield it from the nuclear pull, so less energy is needed to remove it.

(iii) Describe the relationship between group number and the charge of ions formed by elements, including examples from various groups.

For main-group elements, the group number predicts the ion charge, because atoms gain or lose electrons to reach a full outer shell. Group IA elements lose one electron to form +1 ions, such as Na+; group IIA lose two to form +2 ions, such as Mg2+; group VIIA gain one to form -1 ions, such as Cl-; and group VIA gain two to form -2 ions, such as O2-. Group VIIIA noble gases form no ions because their shells are already full.

(iv) Explain the trend in metallic character across a period and down a group in

the periodic table.

Metallic character is the tendency to lose electrons and form positive ions. Across a period it decreases from left to right, because atoms hold their electrons more tightly as nuclear charge grows, so they are less willing to lose them and behave more like non-metals. Down a group it increases, because outer electrons are further from the nucleus and more shielded, so they are lost more easily and the elements behave more strongly as metals.

(v) Discuss the electronic configuration of alkali metals and explain how it contributes to their reactivity.

Alkali metals have the configuration ns1, a single electron in their outer s subshell, on top of a complete noble gas core: lithium is 2,1, sodium is 2,8,1 and potassium is 2,8,8,1. This single electron is easily lost, giving a stable +1 ion with a full shell underneath, so alkali metals are very reactive and are never found free in nature. Reactivity increases down the group because the outer electron gets further from the nucleus and more shielded, making it easier to lose.

(vi) Discuss the general characteristics of transition elements, highlighting their variable oxidation states and catalytic properties.

Transition elements are metals found in the middle of the periodic table, and they are hard, dense, shiny and excellent conductors of heat and electricity. They show variable oxidation states, such as iron as +2 and +3 and copper as +1 and +2, because they can lose different numbers of s and d electrons. They form coloured compounds, such as blue copper(II) sulphate, and they act as catalysts, for example iron in the Haber process and nickel in hydrogenation, because their variable oxidation states let them take part in reactions and return unchanged.

Unit 11: Environmental Chemistry

Exercise 11

A. Multiple choice questions. (Correct option marked with a green tick.)

(1) The primary function of the atmosphere is:

(a) absorbing cosmic rays

(b) generating heat

(c) sustaining life on Earth ✓

(d) producing greenhouse gases

(2) The gas considered a major contributor to global warming is:

(a) nitrous oxide

(b) carbon monoxide

(c) carbon dioxide ✓

(d) sulphur dioxide

(3) The main source of methane emissions is:

(a) Wetlands and rice paddies

(b) Agriculture and livestock farming ✓

(c) Volcanoes and forest fires

(d) Industrial waste gases

(4) The main gas responsible for trapping infrared radiation and contributing to global warming is:

(a) methane

(b) carbon dioxide ✓

(c) nitrous oxide

(d) ozone

(5) The main product formed when sulphur dioxide reacts with water vapour in the atmosphere is:

(a) sulphuric acid ✓

(b) nitric acid

(c) carbonic acid

(d) hydrochloric acid

(6) The poisonous gas produced by the incomplete combustion of carbon-based fuels is:

(a) carbon monoxide ✓

(b) sulphur dioxide

(c) methane

(d) nitrogen dioxide

(7) The major component of Earth's atmosphere is:

(a) oxygen

(b) nitrogen ✓

(c) carbon Dioxide

(d) argon

(8) The percentage of Earth's water found in seas and oceans is:

(a) 71%

(b) 97.2% ✓

(c) 2.15%

(d) 0.63%

(9) The test commonly used in chemistry laboratories to detect the presence of water is:

(a) boiling Point Test

(b) melting Point Test

(c) anhydrous Copper(II) Sulphate Test ✓

(d) coagulation Test

(10) The harmful substance associated with water pollution from fertilisers and detergents is:

(a) dissolved oxygen

(b) harmful microbes

(c) nitrates and phosphates ✓

(d) beneficial mineral

B. Restricted Response Questions (RRQs).

(i) Write the balanced chemical equation for the formation of carbon dioxide from the combustion of methane.

The complete combustion of methane in oxygen gives carbon dioxide and water: CH4 + 2O2 -> CO2 + 2H2O. It is balanced with one carbon atom, four hydrogen atoms and four oxygen atoms on each side.

(ii) Which oxide of nitrogen is reddish-brown and has a pungent smell?

Nitrogen dioxide, NO2, is the reddish-brown oxide of nitrogen with a pungent, choking smell. It forms in vehicle exhausts and lightning, and it is a major air pollutant that contributes to acid rain and smog.

(iii) List four harmful effects of acid rain on the environment.

First, acid rain damages forests by leaching nutrients from soil and harming leaves. Second, it acidifies lakes and rivers, killing fish and other aquatic life. Third, it corrodes buildings, statues and metal structures, especially limestone and marble. Fourth, it damages crops and reduces soil fertility by washing away calcium and magnesium.

(iv) What chemical process leads to the formation of ground-level ozone?

Ground-level ozone forms by photochemical reactions: nitrogen oxides and volatile organic compounds from vehicles and industry react in strong sunlight to produce ozone, O3. Unlike the protective ozone layer high in the atmosphere, ground-level ozone is a harmful pollutant that irritates lungs and damages plants.

(v) Describe the role of sulphur dioxide in the formation of acid rain.

Sulphur dioxide, released by burning coal and oil, reacts with oxygen and water vapour in the atmosphere to form sulphurous acid and then sulphuric acid. These acids fall as acid rain, which damages soil, water, forests and buildings. Sulphur dioxide is therefore one of the two main gases, with nitrogen oxides, behind acid rain.

(vi) Why is distilled water not recommended for long-term human consumption?

Distilled water is not recommended for long-term drinking because distillation removes the dissolved minerals, such as calcium and magnesium, that the body needs and that ordinary drinking water supplies. It also tastes flat, and drinking only distilled water for a long time can disturb the body's mineral balance.

(vii) Why is nitrate leaching a concern for the environment?

Nitrate leaching is a concern because excess nitrate fertiliser washes through soil into groundwater and rivers, polluting drinking water and causing eutrophication. In rivers and lakes, nitrates feed algal blooms that block light and, when the algae die and decay, use up oxygen, killing fish and other aquatic life.

(viii) Which sector consumes over 80% of Pakistan's freshwater resources?

Agriculture consumes over 80% of Pakistan's freshwater resources, mainly for irrigating crops such as wheat, rice, cotton and sugarcane through the Indus canal system. Domestic and industrial use together take only a small share, so efficient irrigation is the key to Pakistan's water security.

C. Extended Response Questions (ERQs).

(i) Describe the composition of clean, dry air and explain the significance of two of its major components.

Clean, dry air is about 78% nitrogen, 21% oxygen, 0.93% argon, 0.04% carbon dioxide and small amounts of other gases. Nitrogen is significant because, although most organisms cannot use it directly, nitrogen-fixing bacteria convert it into compounds that plants need for proteins. Oxygen is significant because it is essential for respiration in animals and plants and for combustion, releasing energy from fuels and food.

(ii) Explain the process of global warming and identify its major causes.

Global warming is the rise in Earth's average temperature caused by greenhouse gases trapping infrared radiation that Earth radiates back towards space. The main causes are burning fossil fuels, which releases carbon dioxide, agriculture and livestock, which release methane and nitrous oxide, deforestation, which removes trees that absorb carbon dioxide, and industrial processes. The trapped heat melts glaciers, raises sea levels and changes

rainfall patterns.

(iii) Identify and explain any four major pollutants found in the atmosphere, and state one effect of each.

Carbon monoxide, from incomplete combustion, is poisonous because it binds to haemoglobin and starves the body of oxygen. Sulphur dioxide, from burning coal and oil, causes acid rain and breathing problems. Nitrogen dioxide, from vehicles, causes smog, acid rain and lung irritation. Particulate matter, tiny dust and soot particles, damages lungs and reduces visibility. Carbon dioxide, though not directly poisonous, drives global warming by trapping heat.

(iv) Explain the significance of water for living organisms and why humans need to consume water daily.

Water is significant because it makes up most of living cells, dissolves nutrients and waste, carries substances in blood and sap, and controls body temperature through sweating. Chemical reactions of life, such as digestion and photosynthesis, take place in water. Humans must drink water daily because they constantly lose it in urine, sweat and breath, and even small shortages cause dehydration, tiredness, headaches and, in severe cases, organ failure.

(v) Outline the key stages involved in the treatment of domestic water and explain the purpose of each stage.

First, pretreatment screens out large debris. Second, flocculation adds chemicals that clump fine particles together. Third, sedimentation lets the clumps settle out. Fourth, filtration passes water through sand or carbon beds to remove remaining particles. Fifth, carbon adsorption removes tastes, odours and some chemicals. Finally, disinfection, usually with chlorine, kills bacteria and viruses, so the water reaching homes is safe to drink.

(vi) Identify and explain the major causes of water scarcity in Pakistan.

The major causes are rapid population growth, which increases demand for drinking water and food; climate change, which makes rainfall less reliable and melts glaciers faster; inefficient water management, including leaking canals and flood irrigation; water pollution, which makes supplies unusable; and limited storage capacity, so monsoon water runs to the sea instead of being saved for dry months. Together these leave Pakistan short of water for farms, cities and industry.

Unit 12: Organic Chemistry

Exercise 12

A. Multiple choice questions. (Correct option marked with a green tick.)

(1) The compound that is NOT considered organic is:

(a) methane

(b) ethanol

(c) sodium carbonate ✓

(d) acetic acid

(2) The general formula of alkanes is:

(a) CnH2n

(b) CnH2n-2

(c) CnH2n+2 ✓

(d) CnH2n+1OH

(3) The unsaturated hydrocarbon among the following is:

(a) methane

(b) ethane

(c) ethene ✓

(d) propane

(4) The type of hydrocarbon that is C4H6 is:

(a) alkane

(b) alkene

(c) alkyne ✓

(d) aromatic compound

(5) The true statement about homologous series is:

(a) members differ by a -COOH group

(b) all have different functional groups

(c) each member differs by a -CH2- unit ✓

(d) all have the same molecular mass

(6) The cyclic hydrocarbon among the following is:

(a) butane

(b) propane

(c) benzene ✓

(d) methane

(7) The catalyst commonly used in the hydrogenation of alkenes and alkynes is:

(a) iron

(b) nickel ✓

(c) copper

(d) cobalt

(8) The product of the complete combustion of methane in the presence of oxygen is:

(a) CO_2 ✓

(b) CO

(c) CH_4

(d) C_2H_6

(9) The reaction that involves the replacement of hydrogen atoms in alkanes by halogen atoms is:

(a) hydrogenation

(b) dehydration

(c) halogenation ✓

(d) oxidation

(10) The primary product when ethane undergoes complete combustion is:

(a) carbon monoxide and water

(b) carbon dioxide and water ✓

(c) carbon soot and water

(d) ethylene and hydrogen

B. Restricted Response Questions (RRQs).

(i) Name four elements commonly found in organic compounds besides carbon.

Four elements commonly found in organic compounds besides carbon are hydrogen, oxygen, nitrogen and sulphur. Halogens such as chlorine and bromine, and phosphorus, are also common in organic compounds.

(ii) Give one example each of a straight-chain, branched, and cyclic organic compound.

A straight-chain example is n-butane, $CH_3CH_2CH_2CH_3$, with all four carbons in one line. A branched example is 2-methylpropane, with a central carbon joined to three methyl groups. A cyclic example is cyclohexane, C_6H_{12}, with its six carbons joined in a ring.

(iii) What does a structural formula represent?

A structural formula shows how the atoms in a molecule are arranged and bonded to each other, not just how many of each atom are present. For example, the structural formula of ethanol, CH_3CH_2OH, shows a methyl group joined to a CH_2 group joined to an OH group, making clear which atoms are bonded together.

(iv) What is the difference between two consecutive members of a homologous series?

Two consecutive members of a homologous series differ by one methylene group, -CH_2-, which adds one carbon atom and two hydrogen atoms. Their molecular mass therefore

differs by 14, while they share the same functional group and similar chemical properties, with gradually changing physical properties.

(v) Which functional group is common in methanol and ethanol?

The hydroxyl group, -OH, is common to methanol, CH_3OH, and ethanol, C_2H_5OH. Because both contain the -OH group, both belong to the alcohol family and show similar chemical behaviour, such as burning and reacting with sodium.

(vi) Do compounds with the same functional group have similar chemical properties?

Yes, compounds with the same functional group have similar chemical properties, because the functional group is the reactive part of the molecule. For example, all alcohols contain -OH and burn and react with sodium in similar ways, and all carboxylic acids contain -COOH and turn blue litmus red. Their physical properties differ gradually with size, but their characteristic reactions are the same.

(vii) Write the balanced chemical equation for the complete combustion of methane.

The complete combustion of methane gives carbon dioxide and water: $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$. It is balanced with one carbon, four hydrogens and four oxygens on each side, and it releases a large amount of heat.

C. Extended Response Questions (ERQs).

(i) Describe the concept of a homologous series. What are the general characteristics of homologous series?

A homologous series is a family of organic compounds with the same functional group and similar structures, in which each member differs from the next by a $-CH_2-$ group. Its general characteristics are: all members share the same functional group and general formula; they have similar chemical properties; their physical properties, such as boiling point, change gradually with size; and each member differs from the next by a molecular mass of 14. Alkanes, alkenes and alcohols are all homologous series.

(ii) What are structural isomers? Give two pairs of examples.

Structural isomers are compounds with the same molecular formula but different arrangements of atoms, giving different structures and properties. Two pairs of examples: butane, $CH_3CH_2CH_2CH_3$, and 2-methylpropane, $CH_3CH(CH_3)CH_3$, both C_4H_{10}; and ethanol, CH_3CH_2OH, and dimethyl ether, CH_3OCH_3, both C_2H_6O. In each pair the atoms are the same in number but joined differently, so the compounds behave differently.

(iii) Compare and contrast alkanes, alkenes, and alkynes in terms of their general formulae, types of bonds, and examples.

Alkanes have the general formula C_nH_{2n+2} and contain only single bonds, for example methane, CH_4, and ethane, C_2H_6; they are saturated and relatively unreactive. Alkenes have C_nH_{2n} and contain at least one carbon-carbon double bond, for example ethene,

C2H4; they are unsaturated and undergo addition reactions. Alkynes have CnH2n-2 and contain at least one carbon-carbon triple bond, for example ethyne, C2H2; they are the most unsaturated and also undergo addition reactions. All three are hydrocarbons used as fuels and chemical raw materials.

(iv) Compare and contrast complete and incomplete combustion of alkanes. Include chemical equations.

Complete combustion happens with plenty of oxygen and gives carbon dioxide and water, releasing maximum heat, for example CH4 + 2O2 -> CO2 + 2H2O. Incomplete combustion happens with limited oxygen and gives carbon monoxide and water, or carbon soot and water, releasing less heat, for example 2CH4 + 3O2 -> 2CO + 4H2O, and CH4 + O2 -> C + 2H2O. Complete combustion is clean and efficient, while incomplete combustion is dangerous because carbon monoxide is poisonous and soot pollutes the air.

(v) Discuss the substitution reaction of alkanes with halogens. What conditions are required for this reaction?

In a substitution reaction, a halogen atom replaces a hydrogen atom in an alkane, for example methane with chlorine gives chloromethane: CH4 + Cl2 -> CH3Cl + HCl. The reaction needs ultraviolet light, which splits chlorine molecules into reactive chlorine atoms that start a free-radical chain reaction, and it continues stepwise to give dichloromethane, trichloromethane and tetrachloromethane. Without light, alkanes do not react with halogens at room temperature, because their C-H bonds are strong and non-polar.

Unit 13: Biochemistry

Exercise 13

A. Multiple choice questions. (Correct option marked with a green tick.)

(1) The elements present in carbohydrates in a 1:2:1 ratio are:

(a) carbon, hydrogen, oxygen ✓

(b) carbon, nitrogen, oxygen

(c) hydrogen, oxygen, nitrogen

(d) carbon, hydrogen, nitrogen

(2) The building blocks of proteins are:

(a) monosaccharides

(b) fatty acids

(c) amino acids ✓

(d) nucleotides

(3) The element present in proteins but absent in carbohydrates is:

(a) carbon

(b) hydrogen

(c) nitrogen ✓

(d) oxygen

(4) The protein responsible for oxygen transport in the blood is:

(a) myoglobin

(b) haemoglobin ✓

(c) keratin

(d) collagen

(5) Essential amino acids are those that:

(a) the body can synthesise

(b) are only found in plants

(c) must be obtained through diet ✓

(d) are non-proteinogenic

(6) Lipids are formed by the combination of:

(a) amino acids and nitrogen

(b) fatty acids and glycerol ✓

(c) monosaccharides and glucose

(d) nucleotides and phosphate

(7) A characteristic of lipids is:

(a) soluble in water

(b) composed of amino acids

(c) provide long-term energy storage ✓

(d) formed by nucleotide chains

(8) The components of a triglyceride molecule are:

(a) three fatty acids and one glycerol ✓

(b) two glycerols and one fatty acid

(c) three amino acids and one glycerol

(d) one phosphate group and two fatty acids

(9) The nitrogenous base found in RNA but not in DNA is:

(a) thymine

(b) adenine

(c) uracil ✓

(d) cytosine

(10) The primary function of DNA in cells is:

(a) energy storage

(b) protein synthesis

(c) genetic information storage ✓

(d) catalysing reactions

B. Restricted Response Questions (RRQs).

(i) What are the basic units (monomers) of carbohydrates?

The basic units of carbohydrates are monosaccharides, simple sugars such as glucose and fructose. Two monosaccharides join to form a disaccharide, such as sucrose, and many join to form a polysaccharide, such as starch or cellulose.

(ii) Name four food sources rich in protein.

Four food sources rich in protein are meat, eggs, milk and lentils. Fish, poultry, dairy products, beans, nuts and seeds are also good protein sources.

(iii) Name the two types of nucleic acids.

The two types of nucleic acids are DNA, deoxyribonucleic acid, and RNA, ribonucleic acid. DNA is double-stranded and stores genetic instructions, while RNA is single-stranded and carries out instructions, including making proteins.

(iv) Differentiate between fat and oil.

Fats and oils are both lipids made of fatty acids and glycerol, but fats are solid at room temperature and usually come from animals, such as butter and ghee, while oils are liquid at room temperature and usually come from plants, such as olive oil and sunflower oil. Fats contain more saturated fatty acids, while oils contain more unsaturated fatty acids.

(v) What is the recommended daily intake of carbohydrates, proteins, and lipids?

According to Table 13.2 of the book, for adults the recommended daily intake is: carbohydrates 45-65% of total calories, or 225-325 g per day; proteins 10-15% of total calories, or 46-56 g per day; and lipids 20-35% of total calories, or 45-80 g per day.

Individual needs vary with age, sex, activity level and health.

(vi) What is the function of enzymes in the body?

Enzymes are protein catalysts that speed up chemical reactions in the body without being used up themselves. They digest food, for example amylase breaks down starch, they build new molecules, and they control metabolism, each enzyme working on a specific reaction under body conditions.

C. Extended Response Questions (ERQs).

(i) Explain the functions of carbohydrates. Give two examples each of monosaccharides, disaccharides, and polysaccharides.

Carbohydrates are the body's main and quickest source of energy, they store energy, and they build structures such as plant cell walls. Monosaccharides: glucose, the blood sugar used directly for energy, and fructose, the sugar in fruit. Disaccharides: sucrose, common table sugar, and lactose, the sugar in milk. Polysaccharides: starch, the energy store in plants such as potatoes and rice, and cellulose, the fibre that forms plant cell walls.

(ii) Describe the structure and dietary sources of lipids.

Lipids are built from fatty acids joined to glycerol; a triglyceride has three fatty acids and one glycerol molecule. They are insoluble in water, store more energy per gram than carbohydrates, insulate organs, protect nerves and form cell membranes. Dietary sources include fatty fish, nuts, seeds, vegetable oils, avocados, butter and ghee, with animal sources giving fats and plant sources giving oils.

(iii) What are nucleic acids? How do DNA and RNA differ in structure and function?

Nucleic acids are large biomolecules made of nucleotide chains that store and transmit genetic information. DNA is double-stranded, shaped as a double helix, contains the sugar deoxyribose and the base thymine, and stays mainly in the nucleus as the permanent instruction book of the cell. RNA is single-stranded, contains ribose and uracil instead of thymine, and carries instructions from DNA to the ribosomes, where proteins are made, and also performs other jobs in the cell.

Unit 14: Empirical Data Collection and Analysis

Exercise 14

A. Multiple choice questions. (Correct option marked with a green tick.)

(1) The prefix milli- in the SI system represents:

(a) one-tenth

(b) one-hundredth

(c) one-thousandth ✓

(d) ten thousand

(2) The SI unit of time is:

(a) minute

(b) hour

(c) second ✓

(d) millisecond

(3) The prefix that represents a factor of 0.001 in the SI system is:

(a) kilo

(b) deci

(c) milli ✓

(d) mega

(4) The meaning of the prefix "centi-" in centimetres is:

(a) 10

(b) 0.1

(c) 0.01 ✓

(d) 100

(5) The instrument commonly used to measure small masses is:

(a) thermometer

(b) stopwatch

(c) analytical balance ✓

(d) measuring cylinder

(6) The standard form A x 10^n represents:

(a) a large number

(b) a small number

(c) both large and small numbers ✓

(d) neither large nor small numbers

(7) The instrument most suitable for measuring the precise volume of a liquid in a titration is:

(a) beaker

(b) graduated cylinder

(c) burette ✓

(d) volumetric flask

(8) The SI unit used to measure the amount of substance is:

(a) mole ✓

(b) kilogram

(c) ampere

(d) litre

(9) The main purpose of using a burette in a laboratory is to:

(a) hold liquids

(b) measure temperature

(c) mix solutions

(d) deliver precise volumes during titration ✓

(10) The way exponents are handled in the multiplication of values in standard form is:

(a) added ✓

(b) subtracted

(c) multiplied

(d) divided

B. Restricted Response Questions (RRQs).

(i) Why are standardized units essential in chemistry?

Standardized units are essential because they let every chemist mean exactly the same thing by a measurement, so results can be compared, repeated and trusted anywhere in the world. Without common units, a mass reported in one laboratory could not be checked or used in another, and recipes, medicines and industrial processes would be unsafe.

(ii) List any three SI base units and their corresponding physical quantities.

Three SI base units are: the metre (m) for length; the kilogram (kg) for mass; and the second (s) for time. Others are the mole (mol) for amount of substance, the kelvin (K) for temperature, and the ampere (A) for electric current.

(iii) Can measurements be precise but not accurate? Explain briefly.

Yes, measurements can be precise but not accurate. Precision means repeated measurements agree closely with each other, while accuracy means they are close to the true value. For example, if a faulty balance always reads 5 g too high, repeated weighings will agree closely, so they are precise, but all are wrong, so they are not accurate.

(iv) Express 0.00075 in scientific notation.

Step 1: move the decimal point to the right until one non-zero digit remains before it: 0.00075 becomes 7.5. Step 2: the point moved 4 places to the right, so the exponent is -4. 0.00075 = 7.5 x 10^-4.

(v) Convert 4.5 x 10^6 to standard decimal form.

Step 1: the exponent 6 means move the decimal point 6 places to the right. Step 2: 4.5 becomes 45, then 450, 4500, 45000, 450000, 4500000. 4.5 x 10^6 = 4,500,000.

C. Extended Response Questions (ERQs).

(i) Compare physical and abstract quantities, providing two examples of each.

Physical quantities can be measured directly with instruments and have units, for example length, measured in metres, and mass, measured in kilograms. Abstract quantities are counted or calculated rather than measured with an instrument, and they may have no unit, for example the number of molecules in a sample, or a ratio such as the mole ratio in an equation. Physical quantities describe what can be observed and weighed, while abstract quantities describe counts, ratios and ideas built from measurements.

(ii) Describe how metric prefixes such as kilo-, deci-, and milli- help express scientific quantities. Provide one example of each.

Metric prefixes scale a unit up or down by powers of ten, so very large or small quantities can be written in convenient numbers. Kilo- means 1000 times, for example 1 kilometre = 1000 metres. Deci- means one-tenth, for example 1 decimetre = 0.1 metre. Milli- means one-thousandth, for example 1 millilitre = 0.001 litre, which is much easier to write than 0.001 litre every time. Prefixes keep numbers short and reduce mistakes in calculations.

(iii) Explain the difference between accuracy and precision in a chemical measurement. Illustrate your answer with an example.

Accuracy is how close a measurement is to the true value, while precision is how closely repeated measurements agree with each other. For example, if the true volume is 25.0 cm3 and four titrations give 25.1, 25.0, 25.2 and 24.9 cm3, the results are both accurate and precise. If they give 27.1, 27.0, 27.2 and 26.9 cm3, they are precise because they agree, but not accurate because they are all about 2 cm3 too high, perhaps because the burette was wrongly calibrated.

(iv) Explain why scientific notation is important in chemistry. Convert the following number to scientific notation: (a) 0.0004560mol (b) 4,300,000uL (c) 0.00089g (d) 560cm3

Scientific notation is important because chemistry deals with extremely large numbers, such as Avogadro's number, and extremely small ones, such as atomic masses, which are hard to write and easy to mistake in ordinary form. Writing them as A x 10^n keeps them compact and makes multiplication and division simple, by adding or subtracting exponents. Conversions: (a) 0.0004560 mol = 4.560 x 10^-4 mol. (b) 4,300,000 uL = 4.3 x 10^6 uL. (c) 0.00089 g = 8.9 x 10^-4 g. (d) 560 cm3 = 5.60 x 10^2 cm3.

Unit 15: Separation Techniques

Exercise 15

A. Multiple choice questions. (Correct option marked with a green tick.)

(1) The main component in a solution that provides the medium for dissolving solutes is:

(a) solute

(b) filtrate

(c) solvent ✓

(d) residue

(2) The term for a solution that contains the maximum amount of solute that can dissolve under specific conditions is:

(a) diluted solution

(b) unsaturated solution

(c) saturated solution ✓

(d) concentrated solution

(3) The process of reducing the concentration of a solute in a solution by adding more solvent is:

(a) filtration

(b) dilution ✓

(c) crystallisation

(d) precipitation

(4) The term that describes a solid that forms when a solute comes out of a solution is:

(a) filtrate

(b) residue

(c) precipitate ✓

(d) solute

(5) The method of separation that uses a fractionating column for precise separation of close boiling points is:

(a) simple distillation

(b) filtration

(c) crystallisation

(d) fractional distillation ✓

(6) The term "miscible" in chemistry refers to:

(a) ability of substances to mix ✓

(b) inability of substances to mix

(c) formation of crystals

(d) presence of solute in a solution

(7) The purpose of a filter in the filtration process is to:

(a) retain solid particles ✓

(b) dissolve solutes

(c) increase solubility

(d) separate immiscible substances

(8) The method used for separating oil and water based on their difference in density and ability to mix is:

(a) filtration

(b) crystallisation

(c) decantation ✓

(d) dilution

(9) The purpose of a filter in the filtration process is to:

(a) retain solid particles ✓

(b) dissolve solutes

(c) increase solubility

(d) separate immiscible substances

(10) The term "miscible" in chemistry refers to the:

(a) ability of substances to mix ✓

(b) inability of substances to mix

(c) formation of crystals

(d) presence of solute in a solution

B. Restricted Response Questions (RRQs).

(i) Differentiate between residue and filtrate.

In filtration, the residue is the solid material left behind on the filter paper, while the filtrate is the liquid that passes through the filter. For example, when a sand and water mixture is filtered, sand is the residue and the water collected below is the filtrate.

(ii) Why does sand remain on the filter paper during filtration?

Sand remains on the filter paper because its particles are larger than the tiny holes, or pores, in the paper, so they cannot pass through. The water molecules are small enough to pass, so the sand is trapped as the residue while the water becomes the filtrate.

(iii) Why is simple distillation unsuitable for separating liquids with close boiling points?

Simple distillation is unsuitable for liquids with close boiling points because both liquids boil and condense together over a narrow temperature range, so they cannot be separated in one evaporation and condensation step. Fractional distillation, with its fractionating column giving repeated condensation and evaporation, is needed for such mixtures.

(iv) How do impurities affect the melting point of a substance?

Impurities lower the melting point of a substance and make it melt over a range of temperatures instead of at one sharp point. For example, pure ice melts sharply at 0 C, but salty ice melts below 0 C and over a range, which is why a broad or lowered melting point shows a substance is impure.

(v) What indicates the presence of impurities in boiling point analysis.

In boiling point analysis, impurities are indicated when a liquid boils over a range of temperatures instead of at one sharp, constant temperature, or when its boiling point is higher than the known value for the pure liquid. Pure water boils steadily at 100 C, while salty water boils above 100 C and over a range.

C. Extended Response Questions (ERQs).

(i) Compare and contrast simple distillation and fractional distillation.

Simple distillation boils a mixture and condenses the vapour once, which suits liquids with widely different boiling points, such as separating water from salt solution, or a liquid from a dissolved solid. Fractional distillation uses a fractionating column that gives many cycles of evaporation and condensation, which suits liquids with close boiling points, such as ethanol and water, or the fractions of crude oil. Both rely on differences in boiling point, but fractional distillation gives a much finer separation at the cost of more complex apparatus.

(ii) Explain how filtration and decantation differ as separation techniques. Under what conditions is each method most appropriate? Give an example for each.

Filtration passes a mixture through a filter that traps the solid and lets the liquid through, so it suits an insoluble solid mixed with a liquid, for example sand in water, where the sand stays on the paper. Decantation carefully pours off the top liquid, leaving a settled solid or a heavier liquid behind, so it suits mixtures where the solid has settled, such as muddy water left to stand, or two immiscible liquids such as oil and water. Filtration is faster and more complete for fine solids, while decantation needs no apparatus but loses some liquid and works poorly with fine particles.

(iii) How can melting point and boiling point analysis be used to assess the purity of a substance?

A pure substance melts and boils at sharp, fixed temperatures: pure water freezes at 0 C and boils at 100 C, and a pure solid melts at one temperature. If a sample melts or boils over a range, or at a shifted temperature, it contains impurities, because impurities disrupt the regular arrangement of particles and change the energy needed for the change of state. Measuring these points and comparing them with known values therefore gives a quick, practical test of purity in the laboratory.

Unit 16: Qualitative Analysis

Exercise 16

A. Multiple choice questions. (Correct option marked with a green tick.)

(1) The gas that turns damp red litmus paper blue is:

(a) carbon dioxide

(b) ammonia ✓

(c) chlorine

(d) hydrogen

(2) When carbon dioxide gas is passed through limewater, it:

(a) turns red

(b) turns milky ✓

(c) turns blue

(d) remains clear

(3) The gas that bleaches damp litmus paper is:

(a) hydrogen

(b) oxygen

(c) chlorine ✓

(d) ammonia

(4) The sound produced when hydrogen gas is tested with a burning splint is:

(a) no sound

(b) sizzling

(c) pop sound ✓

(d) crackling

(5) The flame colour produced by sodium ions (Na+) is:

(a) lilac

(b) intense yellow ✓

(c) crimson red

(d) pale green

(6) The main reason for the different flame colours in the flame test is:

(a) different metal cations absorb and emit light at specific wavelengths ✓

(b) the temperature of the flame

(c) the amount of salt used

(d) the colour of the wire loop

(7) The solution that loses its purple colour when sulphur dioxide gas bubbles through it is:

(a) limewater

(b) acidified potassium permanganate ✓

(c) sodium hydroxide solution

(d) dilute hydrochloric acid

(8) The purpose of using a clean Nichrome wire loop in the flame test is to:

(a) increase the flame temperature

(b) avoid contamination and get a pure flame colour ✓

(c) cool the flame

(d) change the colour of the flame

(9) The mobile phase in paper chromatography refers to the:

(a) chromatography paper

(b) solvent that moves up the paper ✓

(c) sample mixture

(d) coloured spots

(10) A chromatogram is best described as:

(a) a locating agent

(b) a sample spot on the baseline

(c) the pattern formed by separated substances on the paper ✓

(d) the container used for the chromatography

B. Restricted Response Questions (RRQs).

(i) What reagent is used to test for carbon dioxide gas?

Limewater, a clear solution of calcium hydroxide, is used to test for carbon dioxide. When carbon dioxide is passed through limewater, it turns milky because insoluble calcium carbonate forms: $CO_2 + Ca(OH)_2 \rightarrow CaCO_3 + H_2O$.

(ii) How does chlorine gas affect damp litmus paper?

Chlorine gas bleaches damp litmus paper to white. In the presence of water, chlorine forms hypochlorous acid, a powerful oxidising and bleaching agent, which destroys the dye in the litmus paper, so both blue and red litmus turn white.

(iii) What is the stationary phase in paper chromatography?

In paper chromatography, the stationary phase is the chromatography paper itself, which stays in place while the solvent moves. The components of the mixture stick to the paper fibres to different extents, which is what separates them as the solvent rises.

(iv) What is meant by the term "solvent front"?

The solvent front is the furthest point reached by the solvent as it rises up the chromatography paper during the experiment. It is marked before the paper dries, because all distance measurements for Rf values are taken from the baseline to the solvent front.

(v) Calculate the Rf value if a component travels 6cm and the solvent front travels 8cm.

Step 1: use the formula: Rf = distance travelled by the component / distance travelled by

the solvent front. Step 2: substitute the values: 6 / 8. Step 3: divide: 6 / 8 = 0.75. The Rf value is 0.75. Rf values have no units and are always between 0 and 1.

C. Extended Response Questions (ERQs).

(i) Describe the reaction between sulphur dioxide gas and acidified potassium permanganate solution. What observation confirms the presence of sulphur dioxide?

Sulphur dioxide is a reducing agent, and when it bubbles through purple acidified potassium permanganate solution, it reduces the purple manganate(VII) ions to almost colourless manganese(II) ions, so the solution is decolourised from purple to colourless. This loss of purple colour confirms the presence of sulphur dioxide, because few other common gases reduce permanganate in this way.

(ii) Describe the principle of paper chromatography and explain how components of a mixture are separated during the process.

Paper chromatography separates the components of a mixture because they differ in how strongly they stick to the stationary paper and how well they dissolve in the moving solvent. A spot of the mixture is placed on the baseline of the paper, and the solvent rises by capillary action, carrying the components with it. A component that dissolves well in the solvent and sticks weakly to the paper travels far, while one that sticks strongly to the paper travels only a short way, so the components spread out into separate spots on the chromatogram.

(iii) Define the term 'Rf value' and describe how it is calculated and used to identify substances in a mixture.

The Rf value, or retention factor, is the ratio of the distance travelled by a component to the distance travelled by the solvent front, both measured from the baseline: Rf = distance of component / distance of solvent front. It has no units and lies between 0 and 1. Because each substance has a characteristic Rf value in a given solvent, an unknown spot can be identified by comparing its Rf value with those of known substances run on the same chromatogram under the same conditions.